

EXPERIMENT NO: 01 - Acquisition of Volatile and Non-Volatile Memory Disk Images Using FTK Imager

DATE:

1. AIM

To perform a complete forensic acquisition and analysis lifecycle by imaging non-volatile storage using AccessData FTK Imager, capturing live volatile memory (RAM) using the WinPmem command-line utility, and analyzing the extracted memory dump using Volatility 3 commands via the PassMark Volatility Workbench.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tools:**
 - AccessData FTK Imager (GUI for non-volatile physical drive imaging)
 - WinPmem (winpmem_mini_x64_rc2.exe) (CLI tool for volatile memory capture)
 - Volatility 3 Framework (vol.exe / PassMark Volatility Workbench V3.0)
- **Hardware:** Evidence PC/Workstation, Target Storage Drive (hp v221w 16GB USB Device).

3. DESCRIPTION

A thorough digital forensic investigation requires collecting both data at rest and data in motion using strict operational procedures.

- **Non-Volatile Memory Acquisition:** AccessData FTK Imager creates mathematically verifiable, bit-by-bit physical images of storage devices (saved as Raw/dd) without altering the original evidence.
- **Volatile Memory (RAM) Acquisition:** Command-line utilities like **WinPmem** load a specialized kernel driver (pme98A2.tmp) to read physical memory directly. Executing this via CMD writes active processes, network connections, and decrypted keys to a raw dump file (.raw).

- **Volatile Memory Analysis (CLI): Volatility 3** is a framework for analyzing RAM dumps. By passing specific parameters and plugins (e.g., -f <image_path> windows.pslist.PsList) into the command line or workbench, investigators can reconstruct the system's live state and identify suspicious processes.

4. PROCEDURE

Part A: Acquiring Volatile Memory (Disk Image) using FTK Imager

1. Launch **FTK Imager** as an Administrator.
2. Click the **Capture Memory** icon on the top toolbar.

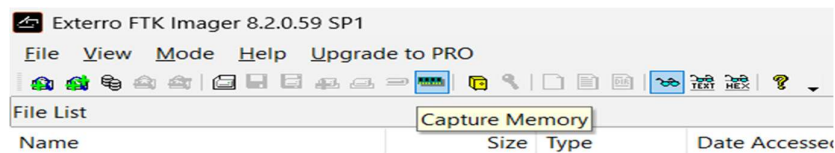


Figure 1: FTK Imager main interface with the Capture Memory option highlighted.

3. Click **Browse** under *Destination path* to select the output folder and type the output file name (e.g., memdump.mem).

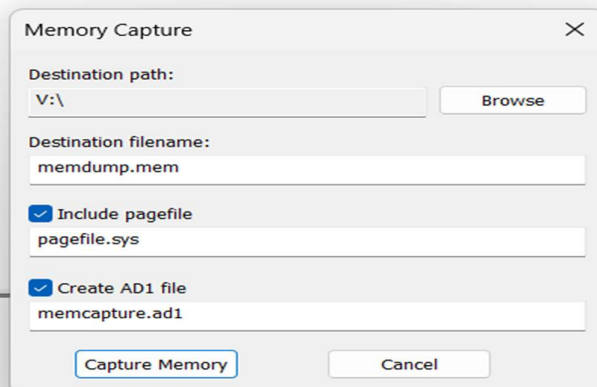


Figure 2: FTK Imager Memory Capture dialog for configuring the destination path and memory dump filename.

4.(Optional) Check **Include pagefile** (pagefile.sys) and **Create AD1 file** if required for the investigation.

5.Click **Capture Memory** to begin acquisition.

6.Wait until the process finishes; the captured memory will be saved with a .mem extension.

Part B: Acquiring Non-Volatile Memory (Disk Image)

1.In FTK Imager, click **File > Create Disk Image** (or click the toolbar icon).

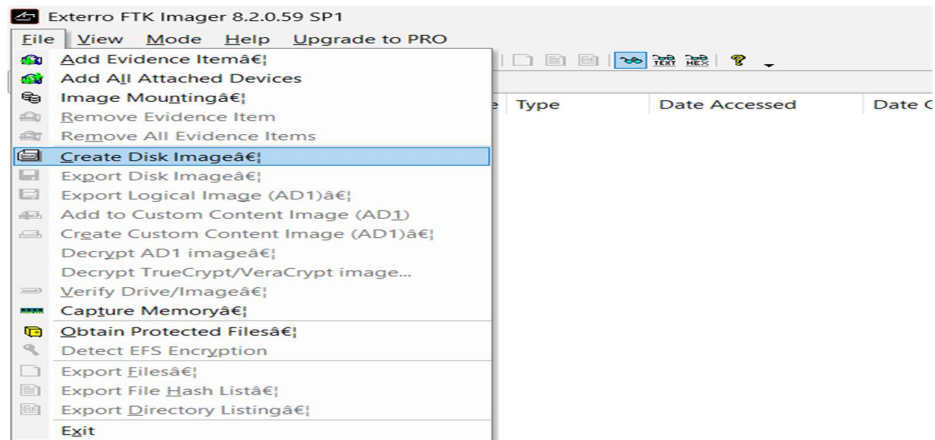


Figure 3: FTK Imager File menu showing the Create Disk Image option.

2.Select the **Source Evidence Type** (e.g., *Physical Drive* or *Logical Drive*) and click **Next**.

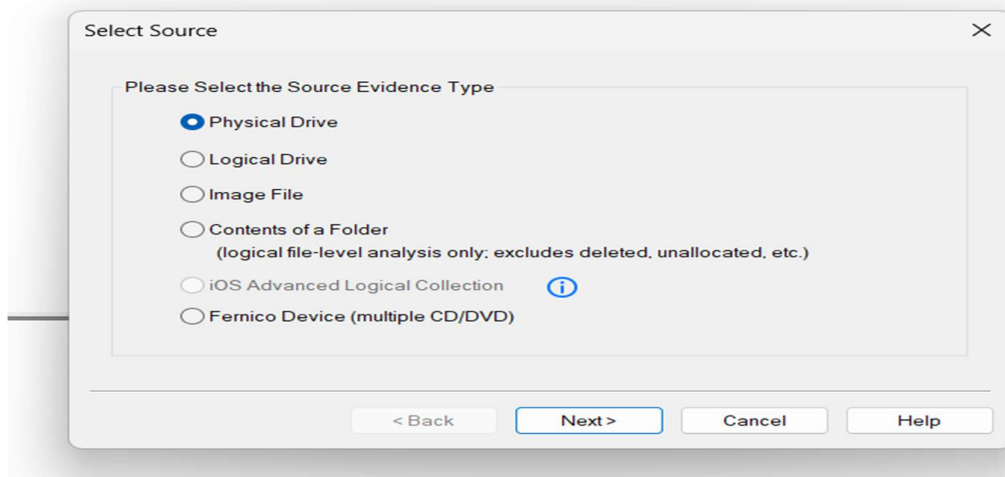


Figure 4: FTK Imager Select Source dialog for choosing the source evidence type, such as *Physical Drive* or *Logical Drive*.

3. Select the target drive from the dropdown list and click **Finish**.

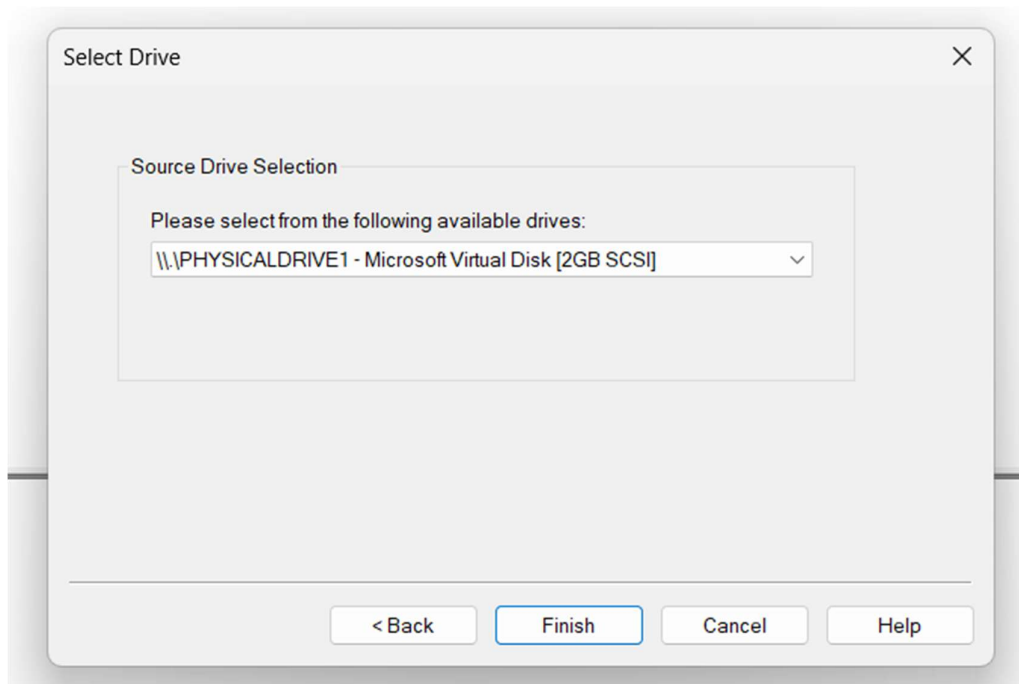


Figure 5: FTK Imager Select Drive dialog displaying the available physical drive selected for imaging.

4. Click **Add...** to configure the image destination:

Select the Image Type (e.g., **Raw (dd)**, **E01**, **SMART**, or **AFF**) and click **Next**.

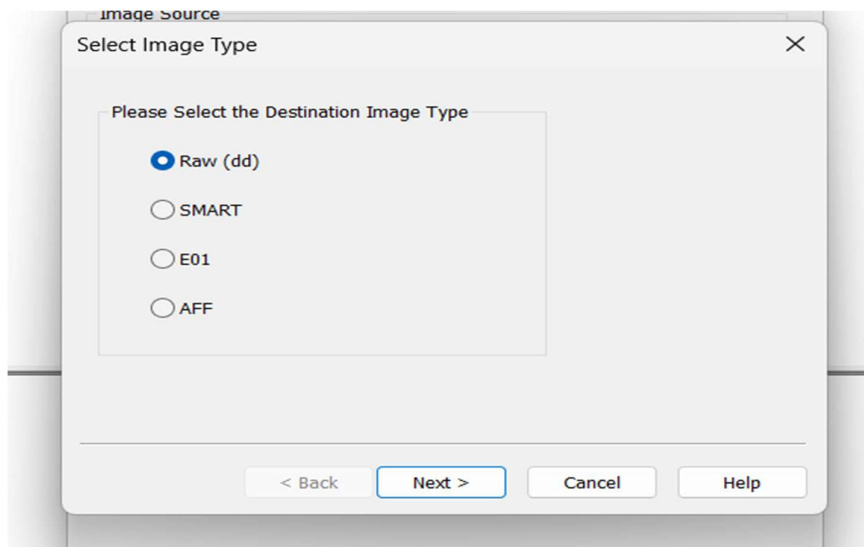
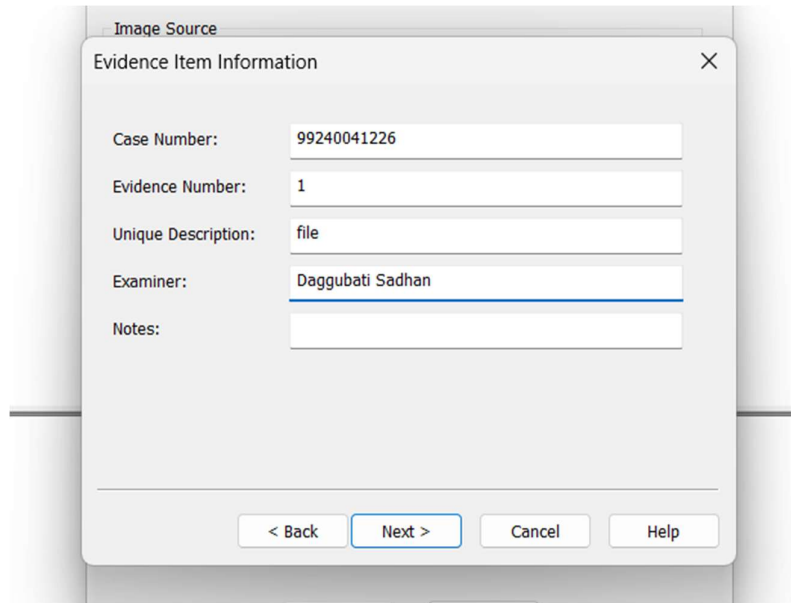


Figure 6: FTK Imager Select Image Type dialog for selecting the disk image format, with Raw (dd) selected.

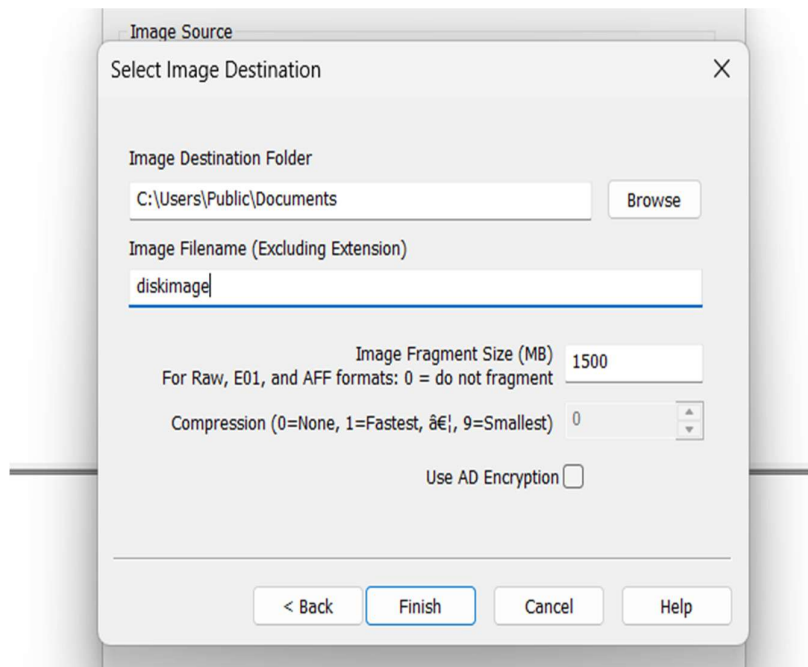
Fill in the *Evidence Item Information* (Case Number, Evidence Number, Examiner Name, Notes) and click **Next**.



The screenshot shows the 'Evidence Item Information' dialog box within the 'Image Source' window. The dialog has a title bar with a close button. It contains five text input fields: 'Case Number' with the value '99240041226', 'Evidence Number' with '1', 'Unique Description' with 'file', 'Examiner' with 'Daggubati Sadhan', and an empty 'Notes' field. At the bottom, there are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'.

Figure 7: FTK Imager Evidence Item Information dialog for entering case number, evidence number, examiner name, and notes.

Specify the **Image Destination Folder** and **Image Filename**.



The screenshot shows the 'Select Image Destination' dialog box within the 'Image Source' window. The dialog has a title bar with a close button. It contains several fields and options: 'Image Destination Folder' with the path 'C:\Users\Public\Documents' and a 'Browse' button; 'Image Filename (Excluding Extension)' with the text 'diskimage'; 'Image Fragment Size (MB)' with a value of '1500' and a note 'For Raw, E01, and AFF formats: 0 = do not fragment'; 'Compression (0=None, 1=Fastest, 8=LZ, 9=Smallest)' with a value of '0' and a spinner control; and a 'Use AD Encryption' checkbox which is unchecked. At the bottom, there are four buttons: '< Back', 'Finish', 'Cancel', and 'Help'.

Figure 8: FTK Imager Select Image Destination dialog for specifying the destination folder, image filename, and image fragment size.

Set **Image Fragment Size (MB)** (set to 0 to keep the file unfragmented as a single image, or leave default values like 1500).

Click **Finish**.

5.Check the box **Verify images after they are created** to ensure hash integrity matching.

6.Click **Start** to begin the disk imaging process.

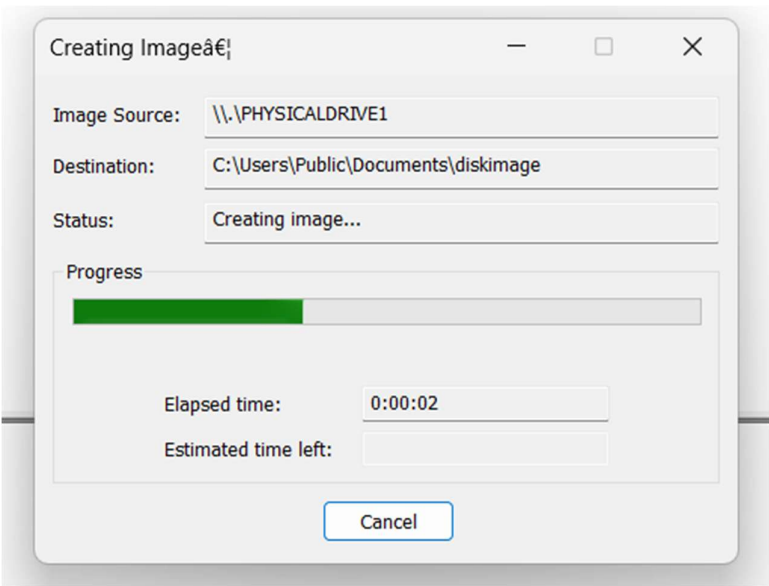


Figure 9: FTK Imager Creating Image dialog showing the progress of the disk imaging process.

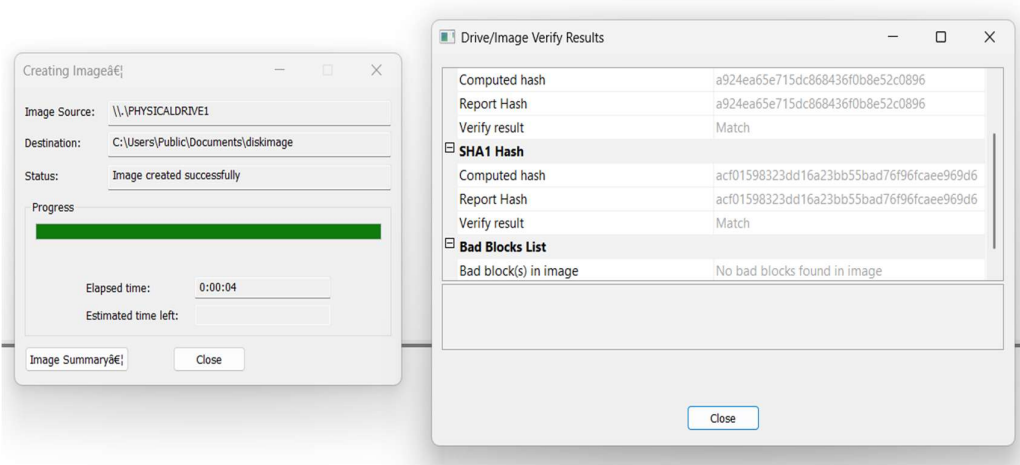


Figure 10: FTK Imager image creation and verification results showing the completed imaging process and hash verification status.

7. Once creation finishes, view the **Image Summary** dialog to confirm that computed MD5 and SHA1 checksums match between the source and created image.

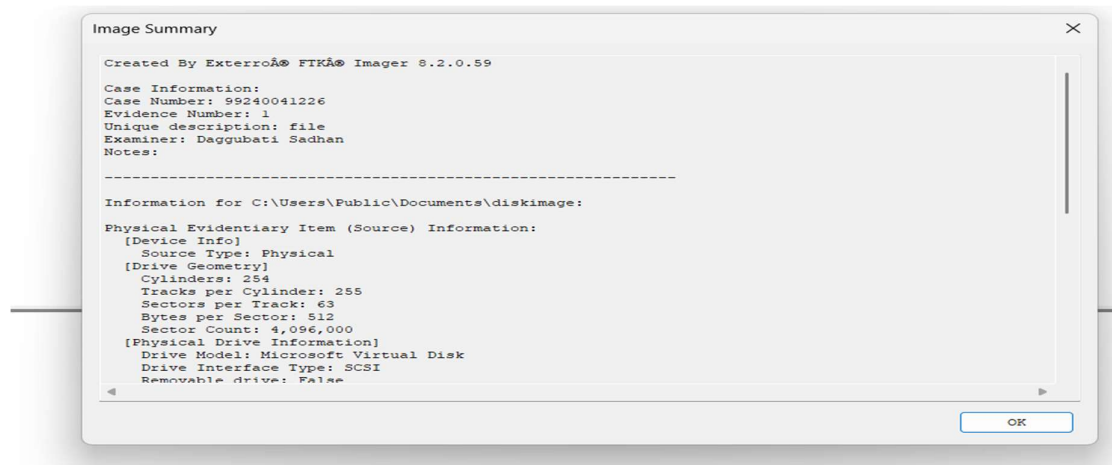


Figure 11: FTK Imager Image Summary dialog displaying details and information about the created disk image.

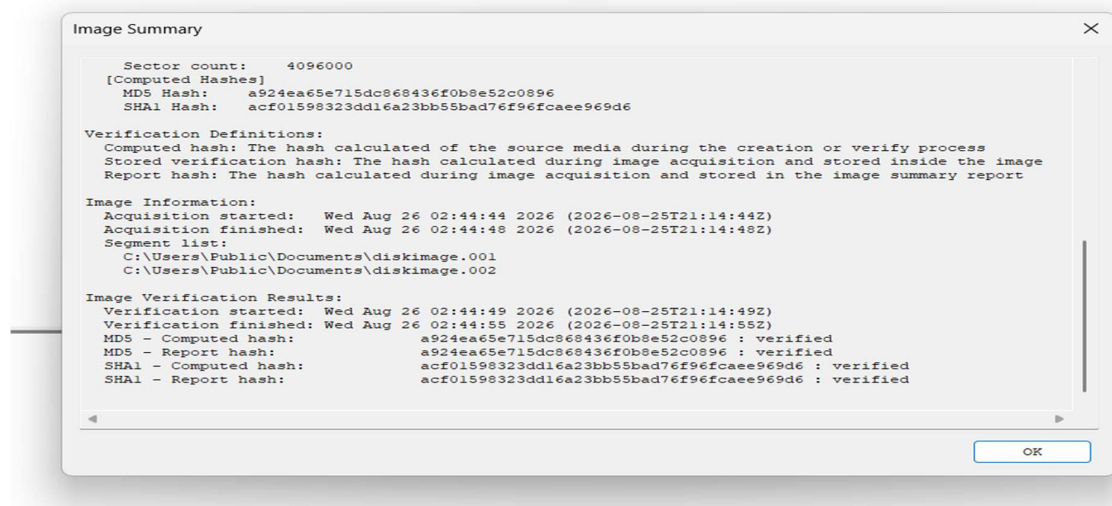


Figure 12: FTK Imager Image Summary showing verification results, including MD5 and SHA1 hash information.

5.RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

6. RESULT

A complete digital forensics acquisition and analysis workflow was successfully executed via a combination of GUI and CLI tools. The target physical USB drive was securely imaged and verified using FTK Imager. Simultaneously, the system's live RAM was extracted via the Command Prompt using WinPmem. Finally, Volatility 3 command-line syntax was utilized to parse the raw memory file, successfully recovering critical live-state artifacts, including the process list and execution timestamps.